

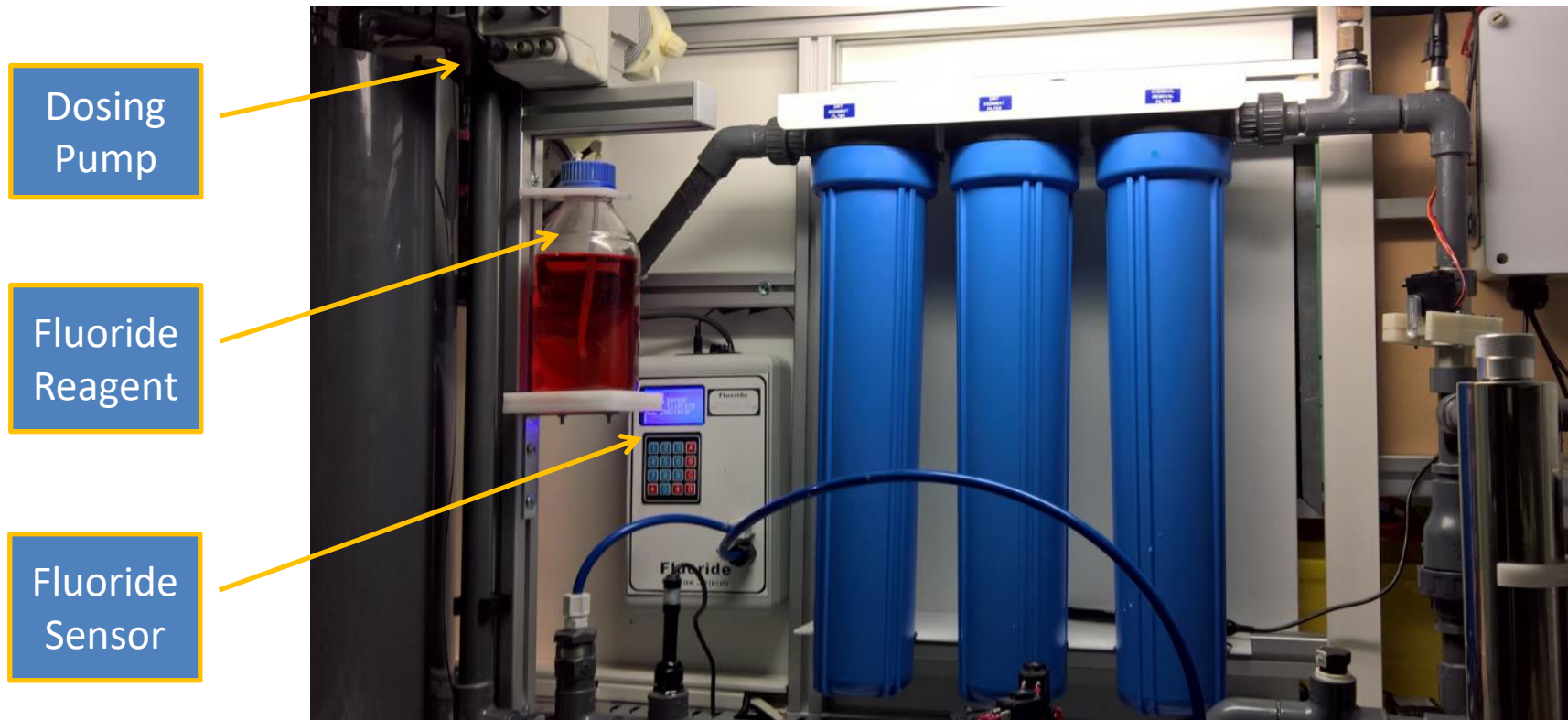
Low Cost Solar Powered Desalination System for Use in Remote Locations with Contaminated Groundwater



Process Cabinet



Pretreatment and Fluoride Sensor

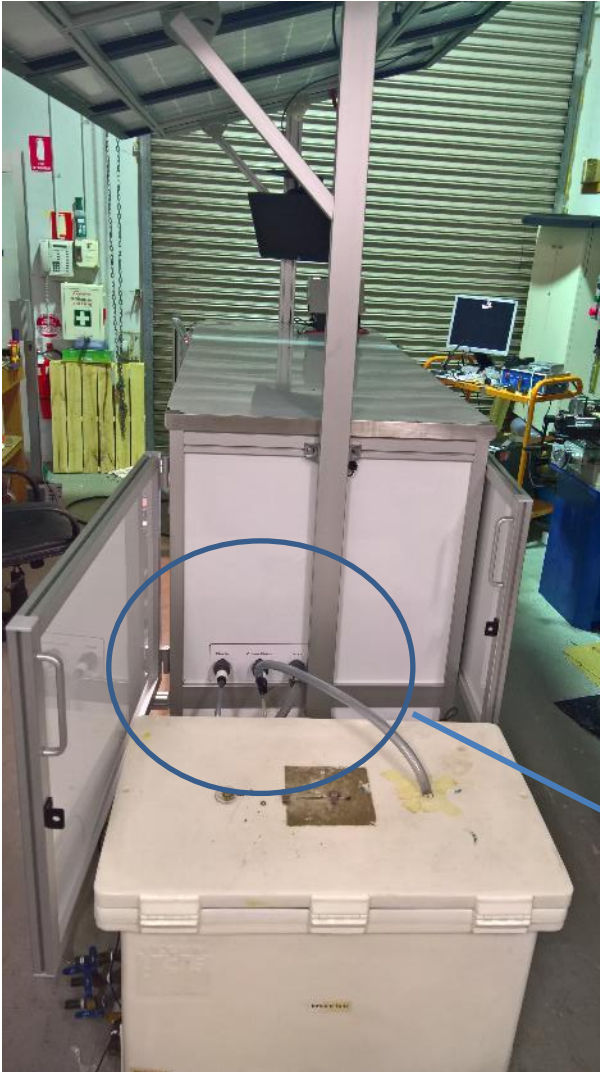


HMI Control System



- System Integration Complete
- Telemetry is functional
- 3G/4G GSM connection is prone to lose contact with server below 15% signal strength. Some improvements required for areas with poor reception.
- Local SQLite database functionality is complete.

Pipe Interface



- $\frac{3}{4}$ " Union couplings fitted for testing period.
- Modifications can be made to suit site requirements.

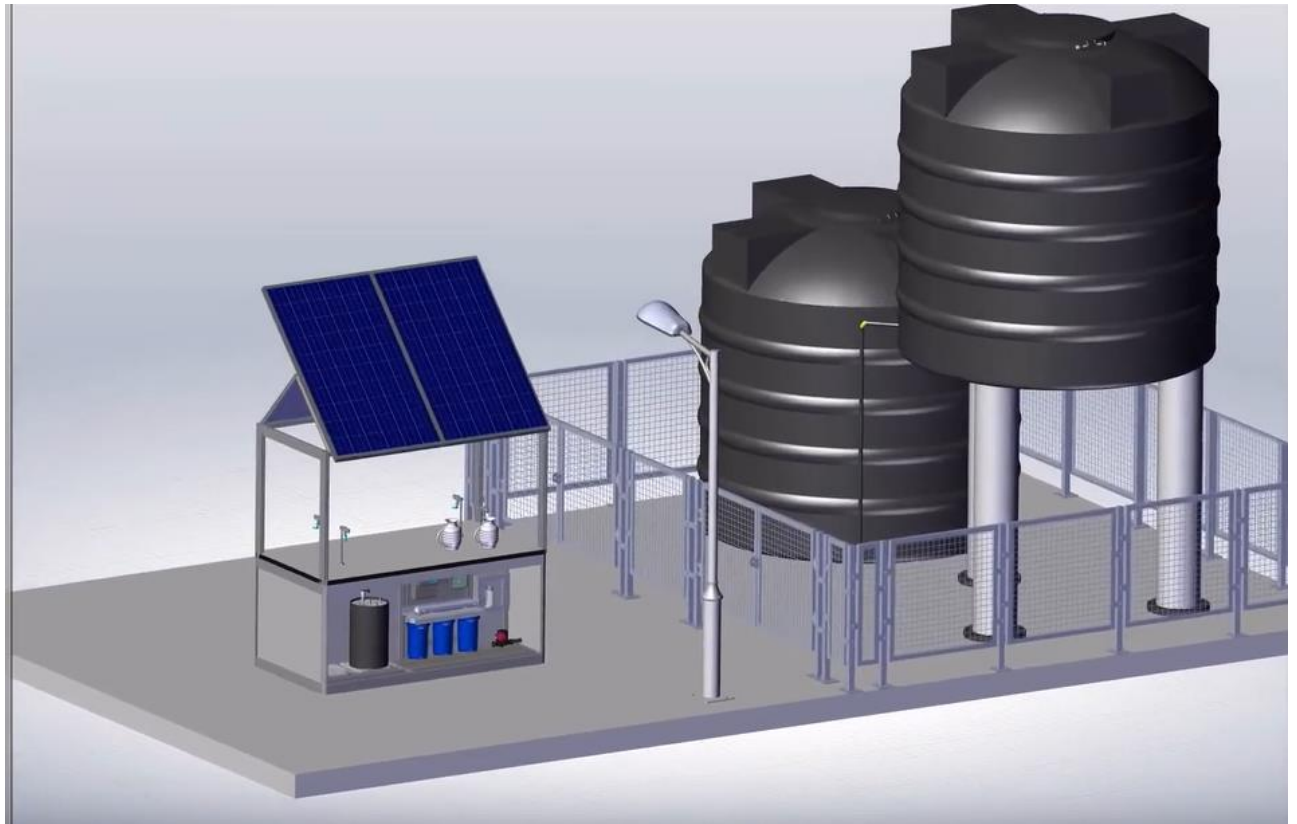


Power Systems Cabinet

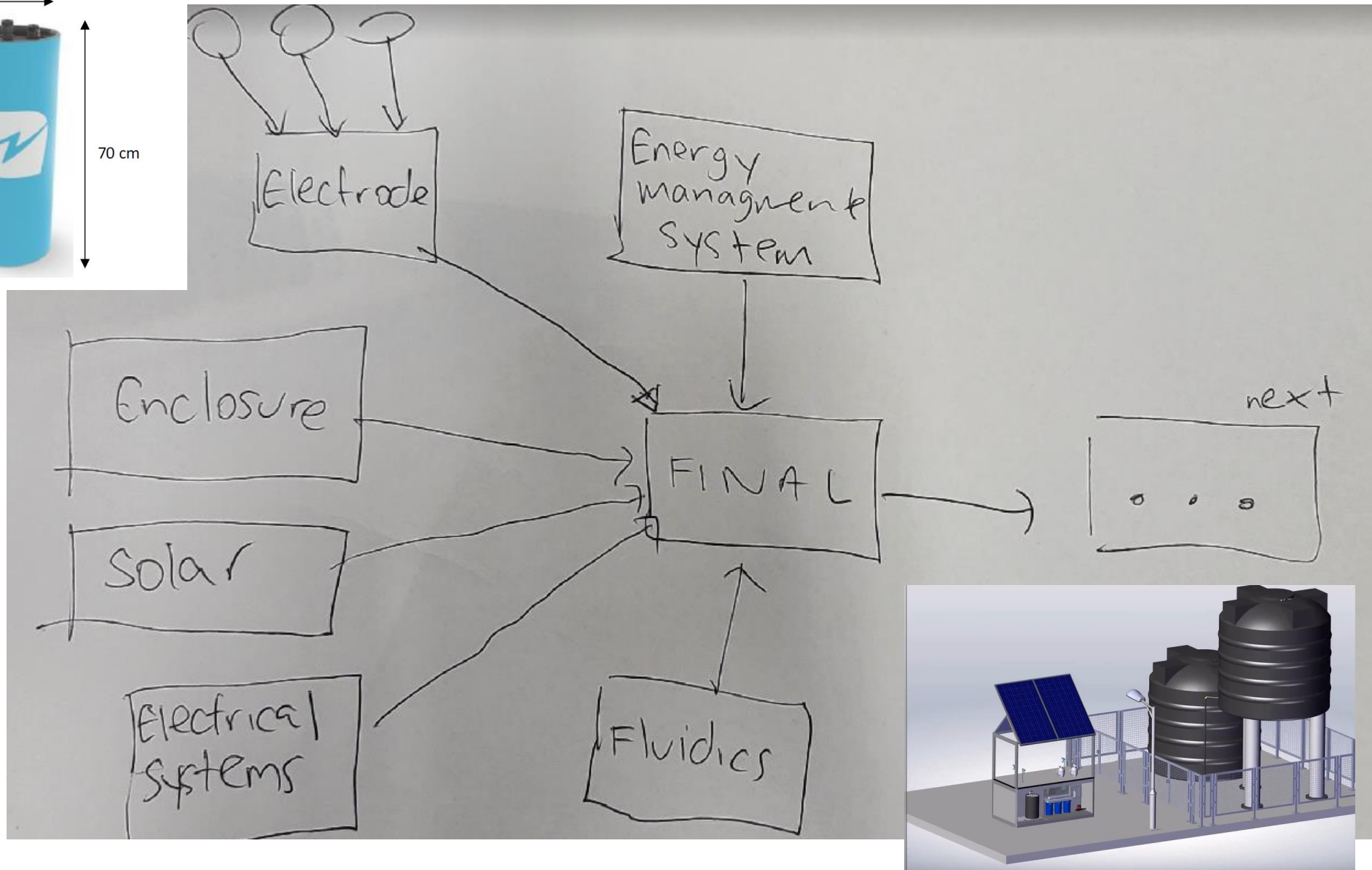
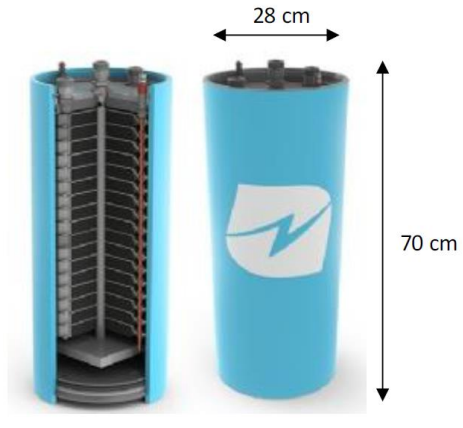


Summary

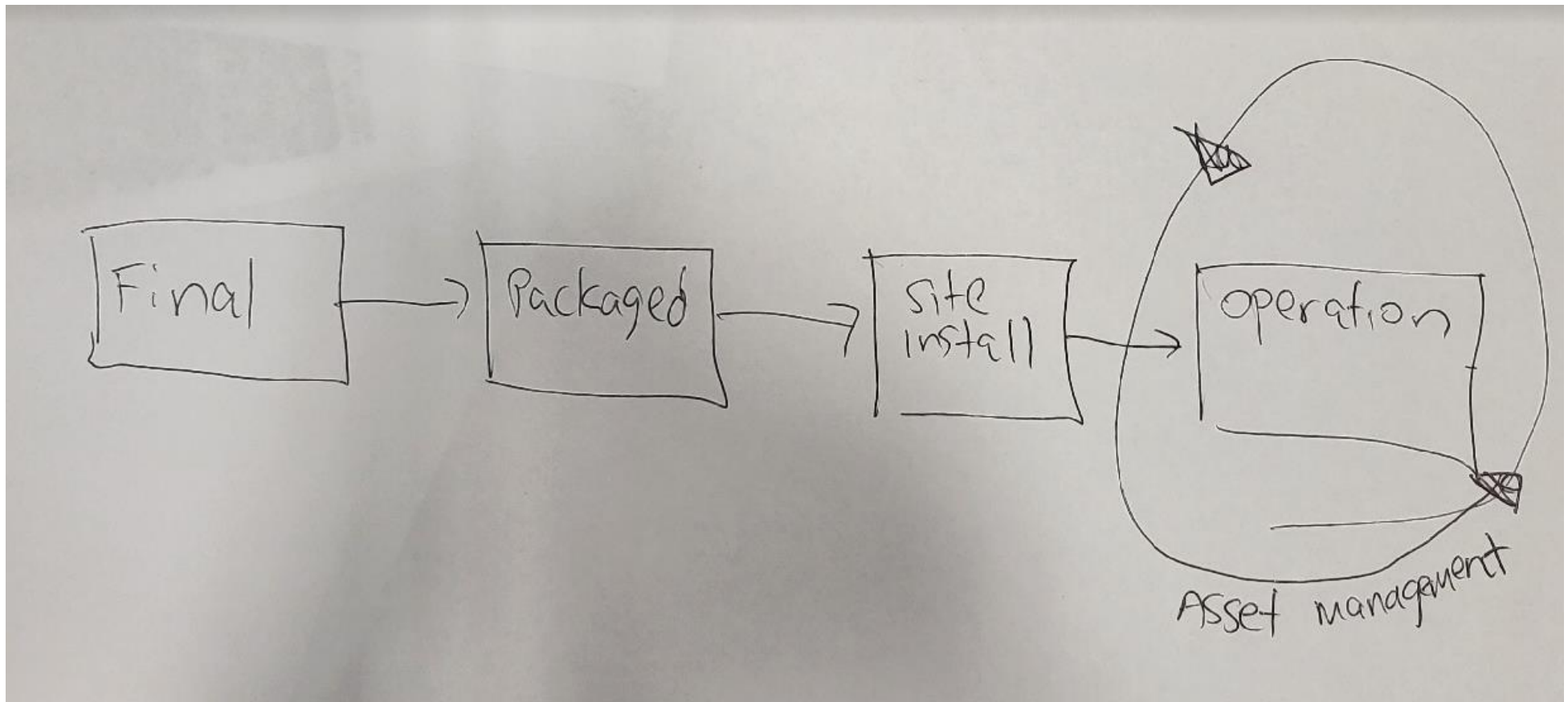
<https://www.youtube.com/watch?v=3D8jdYOLLYw>



Summary



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Operation in the field



Desalination Requirements

- 1000 L/hr
- F: 3.0- $\rightarrow$ 0.5 mg/L
- TDS: 2000 $\rightarrow$ 200 mg/L
- Solar powered
- Easy to operate!
- Easy to maintain!

